



TRAINING AWARENESS

HANDLING OF HEALTHCARE RISK WASTE

Laboratory facility guide for safe segregation, spill response and storage



Presented by Umnotho Hygiene Pty Ltd

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Facility _____ Date _____

Facilitator _____ Session _____

Segregate at the point where waste is generated

Place each item directly into the correct labelled container. Never sort mixed waste by hand. The label and the facility approved container must match the waste stream.

COLOUR	WASTE STREAM	LAB EXAMPLES	CONTAINER OR LABEL
RED	Infectious and infectious anatomical waste	Blood or body-fluid contaminated items, cultures and stocks, tissues, organs and body parts	Biohazard label and correct waste description
YELLOW	Sharps	Needles, lancets, blades, contaminated broken glass, sharp pipettes and puncture hazards	Rigid, leak-resistant and puncture-resistant sharps container
DARK GREEN	Chemical and pharmaceutical waste	Expired medicines, residues, reagents, solvents and chemical products for disposal	Exact contents label; cytotoxic or genotoxic waste needs its own hazard label
BLACK OR FACILITY APPROVED	General non-hazardous waste	Clean packaging, office paper and uncontaminated domestic-type waste	General-waste label and one consistent facility colour
NO PRESCRIBED COLOUR	Radioactive waste	Material contaminated with radionuclides	Radiation symbol and radiation officer details; specialist route only

Five non negotiable rules

- Keep a correctly labelled container within easy reach of every waste-generation point.
- Close containers at the marked fill line or before they reach three-quarters full. Never overfill, compact, shake down or force waste inside.
- Never recap, bend, break or remove a used needle by hand. Dispose of the needle and syringe immediately as one unit where applicable.
- Do not mix chemical, pharmaceutical, cytotoxic, radioactive or infectious waste. Incompatible chemicals can react dangerously.
- If uncertain, stop and ask the laboratory supervisor, infection prevention lead or waste officer before disposal.

Important Check colours against the facility SOP, provincial requirements and contracted waste system. The written hazard label takes priority over colour alone.

Common laboratory waste and the correct container

Use the realistic container photograph together with the written label. Never choose a container by colour alone.

LAB WASTE	CONTAINER	CORRECT ROUTE	SAFE HANDLING
Needles, lancets, blades, contaminated broken glass and sharp pipettes		YELLOW SHARPS	Dispose immediately. Do not recap or detach. Pick up broken glass with forceps, tongs or a scoop.
Blood tubes and specimen containers		RED OR YELLOW CHECK SOP	Breakable or puncture-risk items go to sharps. Intact non-sharp contaminated items follow the infectious route.
Culture plates, swabs, test material and blood-soiled PPE		RED INFECTIOUS	Keep closed. Pre-treat only when required by the approved laboratory SOP.
Human tissue or anatomical material		RED ANATOMICAL	Use a rigid leak-proof container with a sealed lid and approved temperature controls.
Expired medicine and pharmaceutical residue		DARK GREEN PHARMACEUTICAL	Contain liquids and solids securely and apply the correct pharmaceutical label.

Close sharps containers at the fill line. Never reach inside, reopen, shake or transfer the contents.

Specialist laboratory waste

These streams must remain separate from ordinary infectious waste and follow the approved specialist route.

LAB WASTE	CONTAINER	CORRECT ROUTE	SAFE HANDLING
Reagents, solvents, stains and fixatives		COMPATIBLE CHEMICAL CONTAINER	Consult the SDS, record the contents and segregate incompatible chemicals.
Cytotoxic or genotoxic material		DARK GREEN PLUS CYTOTOXIC LABEL	Only trained staff may handle it. Keep it separate from ordinary pharmaceutical waste.
Radionuclide-contaminated material		RADIATION LABELLED CONTAINER	Contact the radiation protection officer. Never place it in ordinary healthcare-risk waste.
Clean wrappers, office paper and uncontaminated packaging		GENERAL WASTE	Use only when the item is not infectious, chemically hazardous, pharmaceutical or radioactive.

Every sealed container needs these details

1 Generator or facility	2 Waste stream	3 Hazard symbol	4 Date sealed	5 Tracking details
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PPE and safe handling

- Wear closed shoes and a laboratory coat or apron. Add gloves, eye and face protection, and respiratory protection according to the risk assessment and SDS.
- Carry a container by its handle or base. Never hold a waste bag against the body, carry a container by the lid, or drag it.
- Inspect the container before movement. If it is damaged or leaking, isolate it and follow the facility overpacking procedure.

Stop isolate assess and prepare

Do not rush into a spill. Identify whether it is biological, chemical, cytotoxic or radioactive before selecting PPE and the spill kit.



Restrict access and keep unprotected people away from the spill

First three actions

1. Stop work, warn others and place a sign or barrier around the spill. Prevent walking through the contaminated area.
2. Assess the substance, amount, splash risk, broken glass, ventilation and exposure. Check the label, facility SOP and Safety Data Sheet where relevant.
3. Put on the required PPE before approaching. Use heavy-duty outer gloves and an apron for routine blood spills; add eye or face protection and closed footwear where splashing is possible.

A biological spill kit should be ready

CONTAINMENT AND CLEANUP	DISPOSAL AND PROTECTION
Absorbent towels or pads; detergent; facility-approved disinfectant; warning sign or barrier	Heavy-duty gloves; disposable apron; eye or face protection; red infectious-waste bag
Forceps, tongs, scraper and scoop for broken glass or sharps	Rigid yellow sharps container and hand-hygiene supplies

Escalate immediately if the spill is unknown, large, volatile, highly infectious, cytotoxic, mercury-containing or radioactive, or if it is beyond your training.

Contain collect clean and disinfect



Cover the spill and use forceps for glass or sharps



Clean first then use the approved disinfectant

Complete the cleanup safely

4. Cover the spill with absorbent towels or pads. Work from the outside edge toward the centre to control spread.
5. Use forceps, tongs, a scraper or scoop for sharp objects and broken glass. Place them directly into the yellow sharps container. Never use your hands.
6. Place used absorbent material in the red infectious-waste container. Clean visible contamination with water and detergent.
7. Apply the facility-approved disinfectant to the cleaned compatible surface. Follow the correct dilution and contact time. National IPC guidance specifies 10,000 ppm available chlorine for blood and body-fluid spills where compatible.
8. Clean reusable equipment, remove PPE safely, perform hand hygiene, report the incident and replace every used spill-kit item.

Chemical cytotoxic and radioactive spills

- Follow the label, SDS and specialist spill procedure. Never add chlorine, water or neutraliser unless the instructions specifically allow it.
- Isolate and call the designated safety, waste or radiation officer when the spill requires specialist control. Package cleanup material in a compatible sealed and fully labelled container.

If a person is exposed

- Needlestick or cut Wash immediately with soap and water. Do not scrub, squeeze or apply bleach.
- Eyes or mucous membranes Flush immediately with clean water or normal saline for at least 15 minutes.
- Remove contaminated clothing, rinse exposed skin, report immediately and obtain urgent occupational-health assessment.

Never mix chlorine with acids, ammonia or other chemicals because toxic gas may be released.

Move and store sealed waste safely



Keep waste streams separated upright closed and clearly labelled



Use a stable easy-to-clean trolley

Safe temporary storage controls

CONTROL	REQUIRED PRACTICE
Security	Access controlled, locked when unattended, clearly marked and restricted to authorised staff.
Condition	Clean, dry, well maintained, pest-free and protected from weather. Surfaces must be washable.
Arrangement	Keep streams separate and containers upright, closed and off the floor. Never reopen, decant or compact sealed waste.
Inspection	Check daily for damage, leakage, odour, overfilling, incorrect labels and overdue collection.
Collection	Use an authorised service provider and retain manifests, collection records and treatment or disposal certificates.

Internal transport rules

- Plan a route away from food, clean supplies, patients and busy public areas.
- Secure containers upright and do not overload the trolley. Clean and disinfect the trolley after use and after any leak.
- Never carry loose or open waste, and never leave a loaded trolley unattended in an unsecured area.

Use the checklist at the start of every shift



Staff must know the spill kit location contents and reporting route

☐	Correct containers are present at each work area and labels face outward.
☐	Sharps containers are secure upright and below the fill line.
☐	Infectious, sharps, chemical, pharmaceutical, cytotoxic, radioactive and general streams are not mixed.
☐	Spill kit is complete and PPE and hand-hygiene supplies are available.
☐	Storage area is locked clean organised and free from leaks or pests.
☐	Sealed containers are dated and ready for documented collection.

Key references

National Department of Health Ideal Clinic Manual version 19 Annexures 53 to 54; National IPC Practical Manual March 2020; Regulations for Hazardous Biological Agents 2022; National Environmental Management Waste Act 59 of 2008; National Norms and Standards for Storage of Waste GN 926 of 2013; Buffalo City Metropolitan Municipality Environmental Health By-law 2010; SANS 10248-1.

This awareness booklet supports training and does not replace the facility SOP, risk assessment, Safety Data Sheets or specialist instructions.

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